

1. What is a Database?

Answer:

A database is an organized collection of data that allows storing, retrieving, updating, and managing information efficiently.

2. What is DBMS?

Answer:

DBMS stands for Database Management System. It is software used to create and manage databases.

Examples:

- MySQL
 - [Oracle Database](#)
 - [PostgreSQL](#)
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3. What is RDBMS?

Answer:

RDBMS stands for Relational Database Management System where data is stored in tables with relationships.

4. What is MySQL?

Answer:

MySQL is an open-source relational database management system used for storing and managing structured data.

5. Difference between DBMS and RDBMS?

Answer:

DBMS	RDBMS
Stores data generally	Stores data in tables
No relationships required	Relationships supported
Less secure	More secure

6. What is a Table in MySQL?

Answer:

A table is a collection of rows and columns used to store related data.

7. What is a Row in Database?

Answer:

A row represents a single record in a table.

8. What is a Column in Database?

Answer:

A column represents a specific attribute or field in a table.

9. What is a Primary Key?

Answer:

A Primary Key uniquely identifies each record in a table and cannot contain NULL values.

10. What is a Foreign Key?

Answer:

A Foreign Key creates relationships between two tables by referencing another table's primary key.

11. What is a Candidate Key?

Answer:

A Candidate Key is a column that can uniquely identify records and can become a primary key.

12. What is Composite Key?

Answer:

A Composite Key consists of two or more columns used together to uniquely identify records.

13. What is Unique Key?

Answer:

A Unique Key ensures all values in a column are unique except NULL values.

14. Difference between Primary Key and Unique Key?

Answer:

Primary Key	Unique Key
Cannot contain NULL	Can contain NULL
Only one per table	Multiple allowed

15. What is NOT NULL constraint?

Answer:

NOT NULL prevents a column from storing NULL values.

16. What is AUTO_INCREMENT?

Answer:

AUTO_INCREMENT automatically generates unique numeric values for records.

17. What is Default Constraint?

Answer:

Default constraint assigns a default value when no value is provided during insertion.

18. What is SQL?

Answer:

SQL stands for Structured Query Language used to communicate with relational databases.

19. Types of SQL Commands?

Answer:

SQL commands are divided into:

- DDL
 - DML
 - DCL
 - TCL
 - DQL
-

20. What is DDL?

Answer:

DDL stands for Data Definition Language used for database structure operations.

Commands:

- CREATE

- ALTER
 - DROP
 - TRUNCATE
-

21. What is DML?

Answer:

DML stands for Data Manipulation Language used for manipulating records.

Commands:

- INSERT
 - UPDATE
 - DELETE
-

22. What is DQL?

Answer:

DQL stands for Data Query Language mainly using the SELECT statement.

23. What is DCL?

Answer:

DCL stands for Data Control Language used for permissions and security.

Commands:

- GRANT
 - REVOKE
-

24. What is TCL?

Answer:

TCL stands for Transaction Control Language used for transaction management.

Commands:

- COMMIT
 - ROLLBACK
 - SAVEPOINT
-

25. What is SELECT query?

Answer:

SELECT retrieves data from database tables.

Example:

```
SELECT * FROM students;
```

26. What is WHERE clause?

Answer:

WHERE filters records based on conditions.

Example:

```
SELECT * FROM students  
WHERE age > 18;
```

27. What is ORDER BY?

Answer:

ORDER BY sorts query results in ascending or descending order.

28. What is GROUP BY?

Answer:

GROUP BY groups rows having the same values for aggregation.

29. What is HAVING clause?

Answer:

HAVING filters grouped data after GROUP BY execution.

30. Difference between WHERE and HAVING?

Answer:

WHERE	HAVING
Filters rows	Filters groups
Used before GROUP BY	Used after GROUP BY

31. What is JOIN in MySQL?

Answer:

JOIN combines rows from multiple tables based on related columns.

32. Types of JOINS?

Answer:

Types of JOINS:

- INNER JOIN
- LEFT JOIN
- RIGHT JOIN
- FULL JOIN
- CROSS JOIN

33. What is INNER JOIN?

Answer:

INNER JOIN returns matching records from both tables.

34. What is LEFT JOIN?

Answer:

LEFT JOIN returns all rows from left table and matching rows from right table.

35. What is RIGHT JOIN?

Answer:

RIGHT JOIN returns all rows from right table and matching rows from left table.

36. What is CROSS JOIN?

Answer:

CROSS JOIN returns Cartesian product of both tables.

37. What is Self Join?

Answer:

Self Join joins a table with itself.

38. What is Normalization?

Answer:

Normalization organizes database tables to reduce redundancy and improve consistency.

39. Types of Normal Forms?

Answer:

Normal forms include:

- 1NF
 - 2NF
 - 3NF
 - BCNF
-

40. What is First Normal Form (1NF)?

Answer:

1NF ensures columns contain atomic values and no repeating groups.

41. What is Second Normal Form (2NF)?

Answer:

2NF removes partial dependency and must already satisfy 1NF.

42. What is Third Normal Form (3NF)?

Answer:

3NF removes transitive dependency and must satisfy 2NF.

43. What is BCNF?

Answer:

BCNF is an advanced normal form where every determinant must be a candidate key.

44. What is Denormalization?

Answer:

Denormalization combines tables to improve read performance.

45. What is Index in MySQL?

Answer:

An index improves data retrieval speed in database queries.

46. Advantages of Indexing?

Answer:

Indexing improves:

- Query speed
 - Search performance
 - Sorting efficiency
-

47. Disadvantages of Indexing?

Answer:

Indexes increase storage usage and slow insert/update operations.

48. What is Clustered Index?

Answer:

Clustered index physically organizes table data based on indexed columns.

49. What is Non-Clustered Index?

Answer:

Non-clustered index stores pointers to actual table rows separately.

50. What is View in MySQL?

Answer:

A View is a virtual table created from SQL queries.

51. What is Stored Procedure?

Answer:

Stored Procedure is a precompiled SQL block stored in database for reuse.

52. What is Trigger in MySQL?

Answer:

A Trigger automatically executes when database events occur.

53. What is Cursor in MySQL?

Answer:

Cursor processes query results row by row.

54. What is Transaction in Database?

Answer:

A transaction is a group of SQL operations executed as a single unit.

55. What are ACID properties?

Answer:

ACID properties ensure reliable database transactions:

- Atomicity
 - Consistency
 - Isolation
 - Durability
-

56. What is Atomicity?

Answer:

Atomicity ensures all transaction operations complete successfully or none execute.

57. What is Consistency?

Answer:

Consistency ensures database remains valid before and after transactions.

58. What is Isolation?

Answer:

Isolation prevents concurrent transactions from interfering with each other.

59. What is Durability?

Answer:

Durability ensures committed data remains permanent even after system failure.

60. What is COMMIT?

Answer:

COMMIT permanently saves transaction changes into database.

61. What is ROLLBACK?

Answer:

ROLLBACK cancels transaction changes before commit.

62. What is SAVEPOINT?

Answer:

SAVEPOINT creates checkpoints inside transactions for partial rollback.

63. What is Deadlock?

Answer:

Deadlock occurs when transactions wait indefinitely for each other's resources.

64. What is Schema in Database?

Answer:

Schema defines database structure including tables, relationships, and constraints.

65. What is Cardinality?

Answer:

Cardinality defines relationship count between tables.

Examples:

- One-to-One
 - One-to-Many
 - Many-to-Many
-

66. What is One-to-One Relationship?

Answer:

One record in first table relates to one record in second table.

67. What is One-to-Many Relationship?

Answer:

One record in first table relates to multiple records in second table.

68. What is Many-to-Many Relationship?

Answer:

Multiple records in one table relate to multiple records in another table.

69. What is Subquery?

Answer:

A subquery is a query written inside another SQL query.

70. What is Correlated Subquery?

Answer:

Correlated subquery depends on outer query values for execution.

71. What is EXISTS operator?

Answer:

EXISTS checks whether subquery returns any records.

72. Difference between DELETE, DROP, and TRUNCATE?

Answer:

DELETE	DROP	TRUNCATE
Removes rows	Removes table	Removes all rows
Can rollback	Cannot rollback	Faster operation

73. What is UNION?

Answer:

UNION combines results from multiple SELECT queries removing duplicates.

74. Difference between UNION and UNION ALL?

Answer:

UNION	UNION ALL
Removes duplicates	Keeps duplicates
Slower	Faster

75. What is Aggregate Function?

Answer:

Aggregate functions perform calculations on multiple rows.

Examples:

- COUNT()
 - SUM()
 - AVG()
 - MAX()
 - MIN()
-

76. What is COUNT() function?

Answer:

COUNT() returns total number of rows.

77. What is AVG() function?

Answer:

AVG() calculates average values from numeric columns.

78. What is SUM() function?

Answer:

SUM() calculates total of numeric column values.

79. What is MAX() function?

Answer:

MAX() returns highest value from a column.

80. What is MIN() function?

Answer:

MIN() returns smallest value from a column.

81. What is Constraint in Database?

Answer:

Constraints enforce rules and restrictions on database columns.

82. What is Referential Integrity?

Answer:

Referential integrity maintains valid relationships between tables.

83. What is Data Redundancy?

Answer:

Data redundancy means duplicate storage of same data unnecessarily.

84. What is Data Integrity?

Answer:

Data integrity ensures data accuracy and consistency in databases.

85. What is Full Backup?

Answer:

Full backup stores complete database data.

86. What is Incremental Backup?

Answer:

Incremental backup stores only changed data after previous backup.

87. What is Database Security?

Answer:

Database security protects data from unauthorized access and attacks.

88. What is SQL Injection?

Answer:

SQL Injection is an attack where malicious SQL code manipulates database queries.

89. How to prevent SQL Injection?

Answer:

SQL Injection can be prevented using:

- Prepared statements
 - Input validation
 - Parameterized queries
-

90. What is Prepared Statement?

Answer:

Prepared statements separate SQL logic from user input improving security.

91. What is Query Optimization?

Answer:

Query optimization improves SQL performance and execution speed.

92. What is Execution Plan?

Answer:

Execution plan shows how database executes SQL queries internally.

93. What is NoSQL Database?

Answer:

NoSQL databases store unstructured or semi-structured data without fixed schemas.

94. Difference between SQL and NoSQL?

Answer:

SQL	NoSQL
Relational	Non-relational
Fixed schema	Flexible schema
Tables	Documents/Key-Value

95. What is MongoDB?

Answer:

MongoDB is a NoSQL database storing data in document format.

96. Advantages of MySQL?

Answer:

MySQL provides:

- Fast performance
 - Reliability
 - Open-source support
 - Security
-

97. What is Database Scalability?

Answer:

Scalability means database can handle increasing data and user load efficiently.

98. What is Replication in MySQL?

Answer:

Replication copies database data from one server to another for availability and backup.

99. What is Sharding?

Answer:

Sharding divides database into smaller parts distributed across multiple servers.

100. Why is MySQL widely used?

Answer:

MySQL is widely used because it is fast, reliable, open-source, secure, and easy to integrate with applications.